Introduction to Proof Systems: First-Order Logic and Natural Deduction

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What is a Proof System?

A **proof system** allows us to produce theorems in a theory. It includes:

- A **formal language** (the language of logical propositions)
  - E.g. propositional logic, N-th order logic
- A **set of axioms**
  - The logical propositions assumed to be true (e.g. $\top$, law of excluded middle, axioms of arithmetic, …)
- A **set of inference rules**
  - Allow us to draw conclusions based on premises
  - Used for proving theorems

First-Order Logic

- **Formal language**
  - A set of variables $\mathcal{X} = \{x, y, z, \ldots\}$
  - A predetermined set of functions with specific arity: Σ
  - A predetermined set of predicates: $\mathcal{P}$
- **Terms:** $t \ ::= \ x \mid f(t_1, \ldots, t_n), \quad f_n \in \Sigma$

- **Formulas:**

$$A, B \ ::= \ P(t_1, \ldots, t_m) \mid A \Rightarrow B \mid A \wedge B \mid A \vee B \mid \top \mid \bot \mid \neg A \mid \forall x, A \mid \exists x, A$$

where $P_m \in \mathcal{P}$

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where $P_m \in \mathcal{P}$

arity

predicate name

Example: Theory of Arithmetic

- Let $\Sigma = \{O_0,\ S_1,\ +_2,\ *_2\}$ (subscripts are arities)
- $\mathcal{P} = \{=_2\}$
- Example formula:

$$\forall x,\ \forall y,\ x + y = y + x$$

Natural Deduction

- A proof system
- It allows us to produce theorems.
- The proof of a theorem is a *derivation tree* produced by applying inference rules

$$\Gamma \vdash A$$

$\Gamma = A_1, \ldots, A_n$	**Hypotheses** (or *context*) (a set of logical propositions)
A	**Conclusion** (a logical proposition)

Natural Deduction: Axioms

$$\frac{}{\Gamma, A, \Gamma' \vdash A}\ \mathsf{AX}$$

The context can include logical propositions we **accept** as true. These are the axioms of the **theory** we represent.

Example: Axioms of Arithmetic

Peano Arithmetic:

$$\forall x,\ \neg(S\,x = O)$$
$$\forall x,\ \forall y,\ S\,x = S\,y \Rightarrow x = y$$

$$\forall x,\ x + 0 = x$$
$$\forall x,\ \forall y,\ x + S\,y = S(x + y)$$

$$\forall x,\ x * 0 = 0$$
$$\forall x\ \forall y,\ x * S\,y = x * y + x$$

$$A(0) \Rightarrow (\forall x,\ A(x) \Rightarrow A(S(x))) \Rightarrow \forall x,\ A(x)$$

$$\forall x,\ x = x$$
$$\forall x\,y,\ x = y \Rightarrow y = x$$
$$\forall x\,y\,z,\ x = y \Rightarrow y = z \Rightarrow x = z$$

Other examples of **first-order** theories: Presburger arithmetic, Zermelo–Fraenkel set theory, Group theory, …

Natural Deduction: Inference Rules

Implication:

$$\frac{\Gamma \vdash A \Rightarrow B \qquad \Gamma \vdash A}{\Gamma \vdash B} \Rightarrow_E \qquad \frac{\Gamma, A \vdash B}{\Gamma \vdash A \Rightarrow B} \Rightarrow_I$$

Conjunction:

$$\frac{\Gamma \vdash A \wedge B}{\Gamma \vdash A} \wedge^l_E \qquad \frac{\Gamma \vdash A \wedge B}{\Gamma \vdash B} \wedge^r_E \qquad \frac{\Gamma \vdash A \qquad \Gamma \vdash B}{\Gamma \vdash A \wedge B} \wedge_I$$

For each logical connective: *introduction rule*, *elimination rule*.

Natural Deduction: Inference Rules

Disjunction:

$$\frac{\Gamma \vdash A \vee B \qquad \Gamma, A \vdash C \qquad \Gamma, B \vdash C}{\Gamma \vdash C} \vee_E \qquad \frac{\Gamma \vdash A}{\Gamma \vdash A \vee B} \vee_I^l \qquad \frac{\Gamma \vdash B}{\Gamma \vdash A \vee B} \vee_I^r$$

Negation:

$$\frac{\Gamma \vdash \neg A \qquad \Gamma \vdash A}{\Gamma \vdash \bot} \neg_E \qquad \frac{\Gamma, A \vdash \bot}{\Gamma \vdash \neg A} \neg_I$$

Natural Deduction: Inference Rules

- **True**

$$\frac{}{\Gamma \vdash \top}\ \top_I$$

- **False**

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash A}\ \bot_E$$

Natural Deduction: Inference Rules

- **True**

Can't eliminate $\top$

$$\frac{}{\Gamma \vdash \top}\ \top_I$$

- **False**

$$\frac{\Gamma \vdash \bot}{\Gamma \vdash A}\ \bot_E$$

Can't introduce $\bot$

Substitution and Free Variables

- **Substitution** $A[t/x]$: replace every free occurrence of variable x in formula A with term t
- **Free variables** $\mathrm{FV}(A)$: the set of variables that appear in A and are not bound by a quantifier ($\forall$ or $\exists$)

  Examples:
  - $\mathrm{FV}(\forall x,\ x + y = y + x) = \{y\}$
  - $\mathrm{FV}(\forall x,\ \forall y,\ x + y = y + x) = \emptyset$

Natural Deduction: Inference Rules

Universal quantifier:

$$\frac{\Gamma \vdash \forall x.\, A}{\Gamma \vdash A[t/x]}\ \forall_E \qquad \frac{\Gamma \vdash A}{\Gamma \vdash \forall x.\, A}\ \forall_I \quad \text{provided } x \notin \mathrm{FV}(\Gamma)$$

Existential quantifier:

$$\frac{\Gamma \vdash \exists x.\, A \qquad \Gamma, A \vdash B}{\Gamma \vdash B}\ \exists_E \quad \text{provided } x \notin \mathrm{FV}(\Gamma) \cup \mathrm{FV}(B) \qquad \frac{\Gamma \vdash A[t/x]}{\Gamma \vdash \exists x.\, A}\ \exists_I$$

- $A[t/x]$: **Substitution** of variable x by term t
- $\mathrm{FV}(\cdot)$: *Free variables*

Natural Deduction: Proof

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{A \wedge B \vdash A \wedge B}\,\mathrm{ax}}{A \wedge B \vdash A}\,\wedge_E^l}{A \wedge B \vdash A \vee B}\,\vee_I^l}{\vdash A \wedge B \Rightarrow A \vee B}\,\Rightarrow_I$$

Natural Deduction

- A **syntactic** way of producing theorems for a first-order theory
- Are the theorems **correct**?
- What does *correct* **mean**?

Natural Deduction

- A **syntactic** way of producing theorems for a first-order theory
- Are the theorems **correct**?
- What does *correct* **mean**?
- A theory described by a set of axioms Γ is **consistent** if

$$\Gamma \nvdash A \wedge \neg A$$

First-Order Logic: Semantics

- **Semantic approach**: assigning meaning to the symbols of the language
- How? Define a *model* $\mathcal{M} = \langle \mathcal{D}, \mathcal{I} \rangle$
  - $\mathcal{D}$: a non-empty set, called the **domain**. Terms map to elements of $\mathcal{D}$.
  - $\mathcal{I}$: *interpretation*
    - For each $f_n \in \Sigma$, a function $[\![f]\!] \in \mathcal{D}_n \to \mathcal{D}$
    - For each $P_n \in \mathcal{P}$, a set $[\![P]\!] \subseteq \mathcal{D}_n$

$$\mathcal{D}_n \overset{\text{def}}{=} \underbrace{\mathcal{D} \times \cdots \times \mathcal{D}}_{n \text{ times}}$$

First-Order Logic: Semantics

- Let $\sigma \in \mathcal{X} \to \mathcal{D}$ be a *variable assignment*
- Define a function assigning values of $\mathcal{D}$ to terms of the language
- How? By **structural recursion** on terms!

$$[\![x]\!]_\sigma = \sigma(x)$$

$$[\![f(t_1, \ldots, t_n)]\!]_\sigma = [f]([\![t_1]\!]_\sigma, \ \ldots, \ [\![t_n]\!]_\sigma)$$

Example: Arithmetic

If $\Sigma = \{O_0,\ S_1,\ +_2,\ *_2\}$, $\quad \mathcal{P} = \{=_2\}$

Define:

$$[O] \stackrel{\text{def}}{=} 0 \qquad\qquad [=] \stackrel{\text{def}}{=} \{(x, x) \mid x \in \mathcal{D}\}$$

$$[S] \stackrel{\text{def}}{=} succ$$

$$[+] \stackrel{\text{def}}{=} +_{\mathbb{Z}}$$

$$[*] \stackrel{\text{def}}{=} *_{\mathbb{Z}}$$

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Different!

The symbol $*$ in the language and the multiplication operation $*_{\mathbb{Z}}$ on $\mathcal{D}$ are different objects.

Example: Arithmetic

Then:

$$[\![S(S(S(S(S\,O))))) * (x + S(S\,O))]\!]_{\{x \mapsto 5\}} =$$

$$[\![S(S(S(S(S\,O)))))]\!]_{\{x \mapsto 5\}} \ *_{\mathbb{Z}} \ \left([\![x]\!]_{\{x \mapsto 5\}} +_{\mathbb{Z}} [\![S(S\,O)]\!]_{\{x \mapsto 5\}}\right) =$$

$$succ(succ(succ(succ(succ(succ(0)))))) \ *_{\mathbb{Z}} \ (5 +_{\mathbb{Z}} succ(succ(0))) = 42$$

First-Order Logic: Semantics

Define a function assigning truth values to formulas. By **structural recursion** on formulas!

$$\begin{aligned}
[\![P(t_1, \ldots, t_m)]\!]_\sigma &= \big([\![t_1]\!]_\sigma,\ \ldots,\ [\![t_n]\!]_\sigma\big) \in [P] \\
[\![\top]\!]_\sigma &= \textbf{true} \\
[\![\bot]\!]_\sigma &= \textbf{false} \\
[\![A \wedge B]\!]_\sigma &= [\![A]\!]_\sigma \ \textbf{and}\ [\![B]\!]_\sigma \\
[\![A \vee B]\!]_\sigma &= [\![A]\!]_\sigma \ \textbf{or}\ [\![B]\!]_\sigma \\
[\![\forall x,\, A]\!]_\sigma &= \textbf{forall}\ v,\ [\![A]\!]_{\sigma[x \mapsto v]} = \textbf{true} \\
[\![\exists x,\, A]\!]_\sigma &= \textbf{exists}\ v,\ [\![A]\!]_{\sigma[x \mapsto v]} = \textbf{true}
\end{aligned}$$

First-Order Logic: Logical Validity

A formula A is **valid** in a model $\mathcal{M}$ when:

- For every $\sigma \in \mathcal{X} \to \mathcal{D}$, $[\![A]\!]_\sigma = \textbf{true}$
- Denoted $\mathcal{M} \models A$

Natural Deduction: Soundness

Let $\Gamma = A_1, \ldots, A_n$

Soundness

If $\Gamma \vdash A$ then for every model $\mathcal{M}$,

$$\mathcal{M} \models \Gamma \implies \mathcal{M} \models A$$

Natural Deduction: Completeness

Let $\Gamma = A_1, \ldots, A_n$

> **Completeness** (Gödel's **Completeness** Theorem)
> If for every $\mathcal{M}$, $\mathcal{M} \models \Gamma \Rightarrow \mathcal{M} \models A$, then $\Gamma \vdash A$.

Equivalently: $\Gamma \vdash A$ or there exists $\mathcal{M}$ such that $\mathcal{M} \models \Gamma, \neg A$

Natural Deduction: Incompleteness

Let $\Gamma = A_1, \ldots, A_n$ be a **consistent** set of propositions capable of describing the axioms of arithmetic.

First Incompleteness Theorem (Gödel)
There exists A such that neither $\Gamma \vdash A$ nor $\Gamma \vdash \neg A$.

Second Incompleteness Theorem (Gödel)
Γ is consistent if for every A, $\Gamma \nvdash A \wedge \neg A$. This can be encoded as a formula in the language itself (Gödel numberings). Let $\mathrm{Con}(\Gamma)$ denote the Gödel encoding of the consistency statement. Then:

$$\Gamma \nvdash \mathrm{Con}(\Gamma)$$

Decidability

- Is there an effective computational procedure that can determine the truth of a formula in first-order logic?
- Some theories are **decidable**
  - E.g. Presburger arithmetic (arithmetic with addition and equality)
- Some theories are **undecidable**
  - E.g. Peano arithmetic
  - Any theory with equality and at least one more binary relation

SMT Solvers

- **SMT** = Satisfiability Modulo Theories
- **Satisfiability problem**: is there a variable assignment that makes A true?
  - Generally **undecidable** but there are decidable subsets
  - A is **unsatisfiable** iff $\neg A$ is **valid** (why?)
- **SMT solvers**:
  - Tools for solving satisfiability for first-order logic
  - Applications: constraint satisfaction, program analysis, program verification
  - Z3, CVC5, Bitwuzla, …
  - Z3 playground: jfmc.github.io/z3-play

Relation to Programming Languages

- **Syntactic analogy** with type systems
  - See *Curry–Howard Isomorphism*
- **Axiomatic semantics** (coming up next)
  - Proof system for program specifications
- Use of **SMT solvers** in **program analysis**

References

- Samuel Mimram, *Program = Proof*, Chapter 5
- Lorenz J. Halbeisen, *First-Order Logic in a Nutshell*
- Melvin Fitting, *First-Order Logic and Automated Theorem Proving*, Chapter 5